

Literature status: the two final mean-ergodicity questions in arXiv:2010.06938

Answers identified in arXiv:2206.00341

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Status: answered in later literature. Questions 3.15 and 3.16 of arXiv:2010.06938 are both settled by Theorems 1.3 and 1.4 of arXiv:2206.00341. In particular, mean and uniform mean ergodicity coincide for every composition operator on $H^\infty(\mathbb{B}_n)$, and a symbol without an interior fixed point never induces a mean-ergodic composition operator.

1 Questions in the source

Let $\varphi: \mathbb{B}_n \rightarrow \mathbb{B}_n$ be holomorphic, let $C_\varphi f = f \circ \varphi$, and, in the interior-fixed-point case, let ρ be the holomorphic retraction associated with φ . Theorem 3.6 of Keshavarzi [1] proves that if, for some $k \in \mathbb{N}$,

$$\sup_{z \in \mathbb{B}_n} \beta(\varphi^{kj}(z), \rho(z)) \longrightarrow 0, \quad (\text{B})$$

then C_φ is uniformly mean ergodic on $H^\infty(\mathbb{B}_n)$. On PDF page 11, the paper then asks:

Question 3.15. Does the inverse [converse] of Theorem 3.6 hold?

Question 3.16. Is it true that composition operators induced by holomorphic functions that have no interior fixed points are not mean ergodic (extending Theorem 3.14)?

2 Exact later answers

Keshavarzi and Hedayatian [2] give the following characterization as Theorem 1.3 on PDF page 2. If φ has an interior fixed point, then the following are equivalent:

1. C_φ is mean ergodic on $H^\infty(\mathbb{B}_n)$;
2. C_φ is uniformly mean ergodic there;
3. for some $k \in \mathbb{N}$, $\|\varphi^{kj} - \rho_\varphi\|_\infty \rightarrow 0$.

This also supplies the *exact* converse requested in Question 3.15. Indeed, uniform mean ergodicity implies mean ergodicity and hence condition (3). In the proof of (3) $\Rightarrow$ (2) (PDF pages 4–6), Theorem 1.2 and the Kobayashi/Bergman estimate convert condition (3) into (B). Together with the source’s Theorem 3.6, this yields

$$C_\varphi \text{ uniformly mean ergodic} \iff (\text{B}) \text{ holds for some } k.$$

Theorem 1.4 of the later paper, also stated on PDF page 2 and proved on PDF pages 10–11, says directly that if φ has no interior fixed point, then C_φ is not mean ergodic on $H^\infty(\mathbb{B}_n)$. The paragraph immediately before the theorem explicitly identifies it as the answer to the source’s Question 3.16.

The proof addresses the fixed-witness obstruction missing from the source’s uniform-ergodicity argument. It restricts the orbit operators to the first coordinate copy of $H^\infty(\mathbb{D})$. Strong convergence there would, using its Grothendieck–Dunford–Pettis structure and an operator-theoretic bounded-below argument, force operator-norm convergence. Boundary Denjoy–Wolff convergence and the usual peak functions then contradict that norm convergence.

3 Scope and provenance

Both concrete questions are answered in full. The supporting authors knew the provenance: their Theorem 1.4 explicitly cites Question 3.16, and their introduction describes Theorem 1.3 as completing the interior-fixed-point case left by the first author’s earlier paper. The broader side question whether $H^\infty(\mathbb{B}_n)$ itself is a Grothendieck space with the Dunford–Pettis property is not answered by these theorems and is not claimed here. This packet records a literature resolution, not a new proof.

References

- [1] H. Keshavarzi, *Mean ergodic composition operators on $H^\infty(\mathbb{B}_n)$* , arXiv:2010.06938; Positivity 26 (2022), article 30, doi:10.1007/s11117-022-00901-5.
- [2] H. Keshavarzi and K. Hedayatian, *Ergodic behaviors of composition operators acting on space of bounded holomorphic functions*, arXiv:2206.00341v1, 2022.